Housing Market Analysis Dashboard Using Tableau and Flask

1. INTRODUCTION

1.1 Project Overview

The Housing Market Analysis Dashboard project aims to analyze housing data to identify trends and factors influencing house prices. The analysis focuses on various housing attributes such as renovation years, house age, number of bedrooms, bathrooms, floors, and basement area.

Interactive visualizations are created to represent patterns in housing sales and property characteristics. These visualizations help stakeholders understand market trends and make informed decisions.

The dashboard provides insights into:

• Total housing sales

• Impact of renovations on property prices

• Distribution of house age

• Relationship between house features and property age

The project also integrates the Tableau dashboard into a web interface using Flask, allowing users to access and interact with the dashboard through a browser.

1.2 Purpose

The purpose of this project is to analyze housing market data and provide meaningful insights through visual analytics.

Objectives:

• Understand factors affecting house prices

• Analyze the impact of renovations on property values

• Visualize housing data using interactive dashboards

• Provide decision-making insights for real estate stakeholders

• Integrate the dashboard into a web application using Flask

2. IDEATION PHASE

2.1 Problem Statement

Real estate companies often struggle to identify patterns in housing sales data due to the large volume of information available. Without proper visualization tools, it becomes difficult to analyze trends related to property age, renovation status, and housing features.

ABC Company requires an interactive dashboard that can analyze housing data and present insights in a clear and understandable manner.

2.2 Empathy Map Canvas

Says:

• Need insights into housing market trends

• Renovation impact should be understood

Thinks:

• Which factors influence house prices?

• Are renovated houses more valuable?

Does:

• Analyze sales data

• Review market reports

• Evaluate property trends

Feels:

• Needs simple tools to analyze data

• Wants visual insights

2.3 Brainstorming

Ideas considered:

• Interactive dashboards

• Histograms for price distribution

• Pie charts for renovation trends

• Bar charts for housing features

• Web integration for accessibility

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User opens the web application

2. User views the housing dashboard

3. User interacts with filters and charts

4. User analyzes insights

5. User makes data-driven decisions

3.2 Solution Requirements

Functional Requirements:

• Load housing dataset

• Create calculated fields

• Build visualizations

• Develop dashboards

• Integrate with Flask

Non-Functional Requirements:

• Fast dashboard loading

• Interactive user interface

• Accurate visualization

• Easy accessibility

3.3 Data Flow Diagram

Housing Dataset

↓

Data Preparation

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Tableau Visualization

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Tableau Dashboard

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Tableau Public / Server

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Flask Web Application

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User Interface

3.4 Technology Stack

Python – Backend programming

Flask – Web framework

Tableau – Data visualization

HTML/CSS – Web interface

Visual Studio Code – Development environment

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution transforms raw housing data into meaningful insights using visualization tools.

4.2 Proposed Solution

• Collect housing dataset

• Prepare data

• Create visualizations

• Develop dashboard

• Publish dashboard

• Embed dashboard using Flask

4.3 Solution Architecture

Data Source → Tableau Dashboard → Tableau Public → Flask Web App → User Browser

5. PROJECT PLANNING & SCHEDULING

Project stages:

1. Data collection

2. Data cleaning

3. Visualization creation

4. Dashboard development

5. Storyboard creation

6. Web integration

7. Testing and deployment

6. FUNCTIONAL AND PERFORMANCE TESTING

Performance testing ensures that the dashboard works efficiently.

Tests performed:

• Dashboard loading time

• Filter functionality

• Visualization accuracy

• Web integration performance

7. RESULTS

The final system provides an interactive housing market dashboard.

Key Insights:

• Recently renovated houses have higher prices

• Older houses dominate the dataset

• Newer houses contain more bathrooms and floors

7.1 Output Screenshots

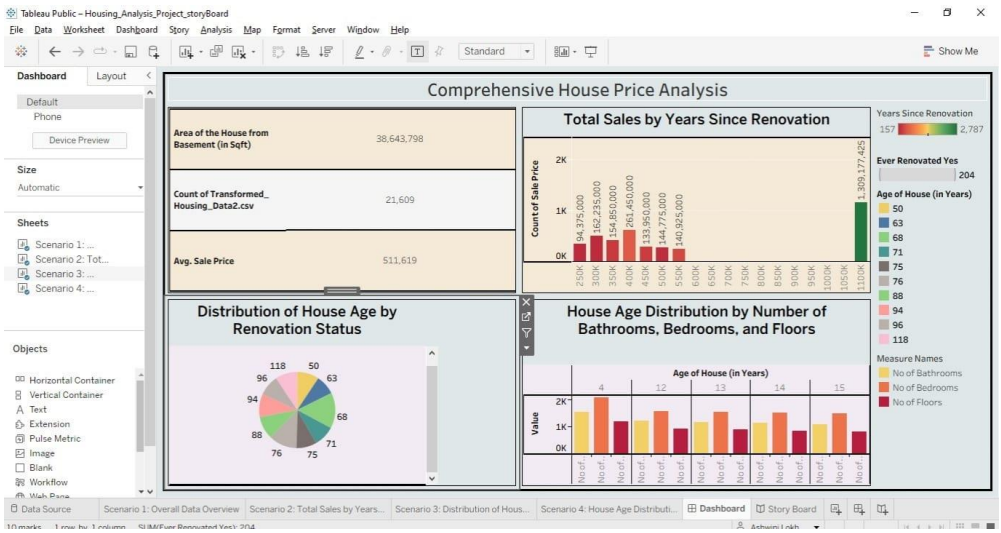
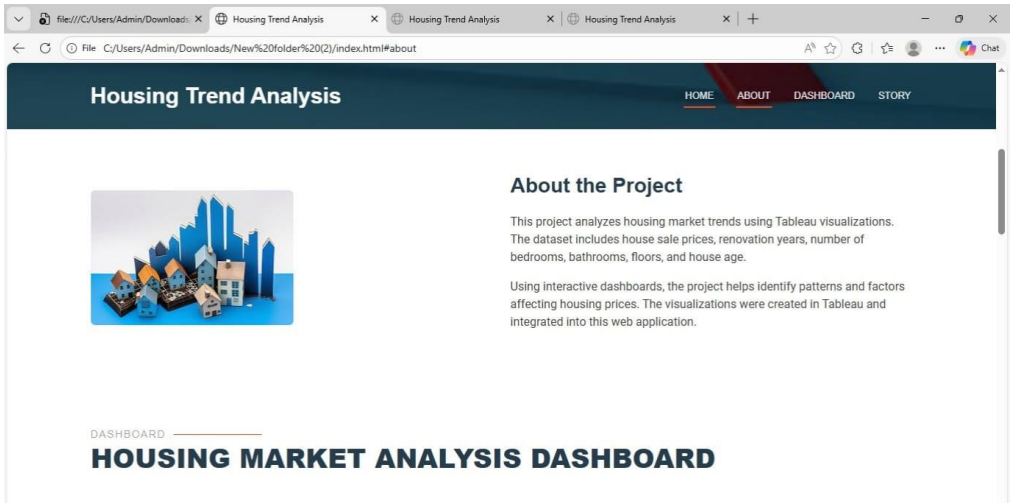
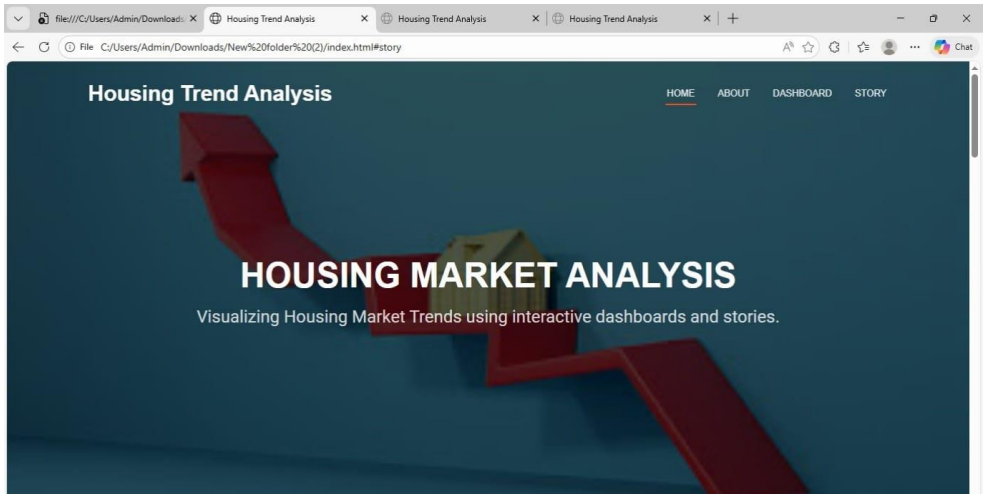


Tableau Public – Housing_Analysis_Project_storyBoard
Comprehensive House Price Analysis
Area of the House from Basement (in Sqft)
38,643,798
Count of Transformed_Housing_Data2.csv
21,609
Avg. Sale Price
511,619
Total Sales by Years Since Renovation
Count of Sale Price
Years Since Renovation
157
2,787
Ever Renovated Yes
204
Age of House (in Years)
Distribution of House Age by Renovation Status
House Age Distribution by Number of Bathrooms, Bedrooms, and Floors
Age of House (in Years)
Value
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors


Housing Trend Analysis
HOME
ABOUT
DASHBOARD
STORY
About the Project
This project analyzes housing market trends using Tableau visualizations. The dataset includes house sale prices, renovation years, number of bedrooms, bathrooms, floors, and house age.
Using interactive dashboards, the project helps identify patterns and factors affecting housing prices. The visualizations were created in Tableau and integrated into this web application.
DASHBOARD
HOUSING MARKET ANALYSIS DASHBOARD


Housing Trend Analysis
HOME
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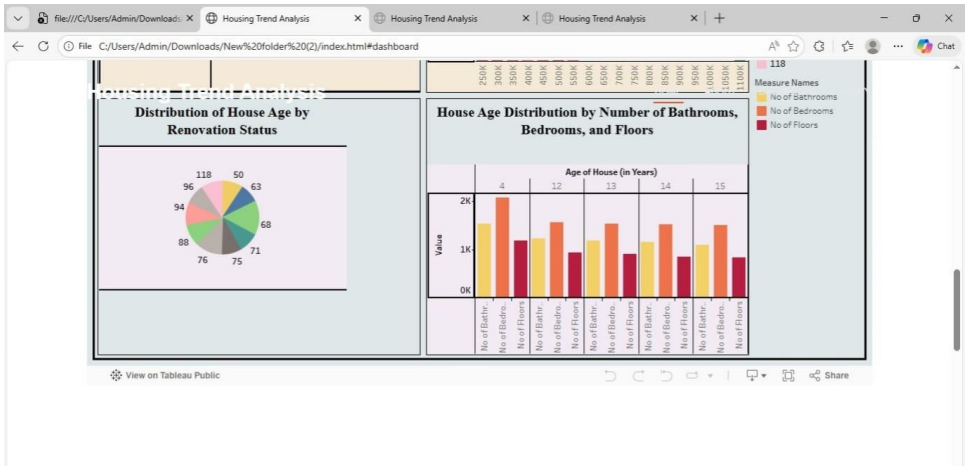
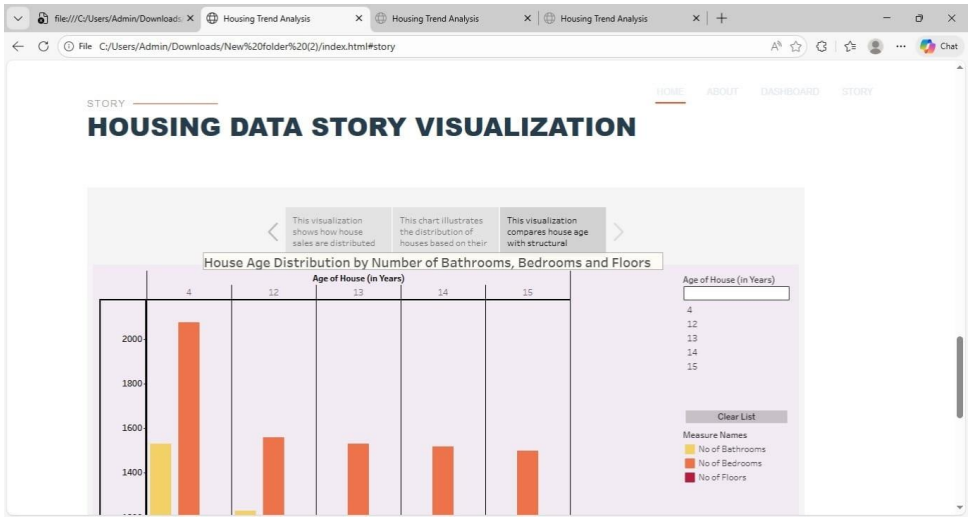
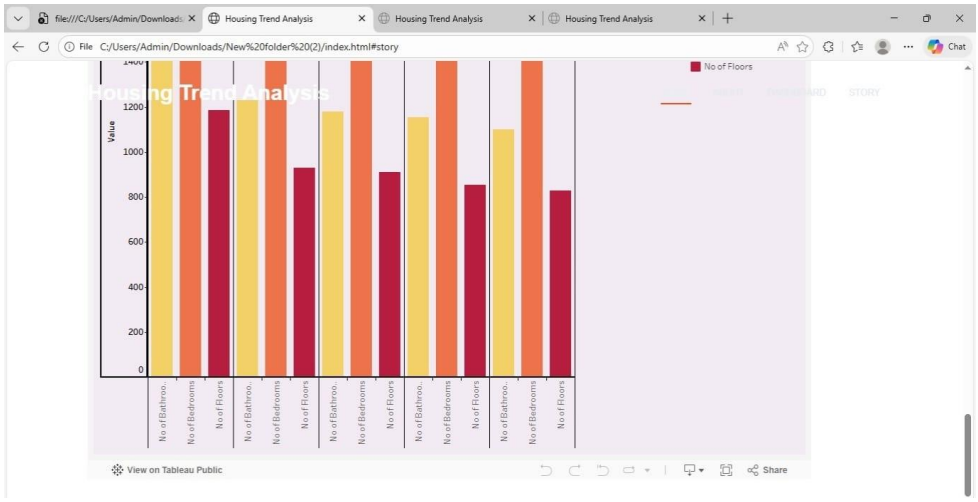
STORY
HOUSING MARKET ANALYSIS
Visualizing Housing Market Trends using interactive dashboards and stories.







8. ADVANTAGES & DISADVANTAGES

Advantages:

• Interactive visualizations

• Easy understanding of housing trends

• Supports data-driven decisions

• Web accessibility

Disadvantages:

• Internet required for Tableau Public

• Large datasets may slow dashboards

9. CONCLUSION

The project successfully analyzes housing data and provides insights into the real estate market using visualization tools. Integration with Flask allows users to interact with the dashboard through a web interface.

10. FUTURE SCOPE

Future improvements may include:

• Machine learning predictions for house prices

• Real-time housing data

• Advanced interactive dashboards

• Mobile-friendly web applications

11. APPENDIX

Source Code (Flask)

```
from flask import Flask, render_template

app = Flask(__name__)


@app.route('/')

def dashboard():
```

```
    return render_template('dashboard.html')


if __name__ == '__main__':
    app.run(debug=True)
```

Dataset: King County House Sales Dataset

GitHub Link: (Add your GitHub repository link)

Tableau Dashboard Link:
https://public.tableau.com/views/Housing_Analysis_Project/Dashboard?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link